

Course Code	Name of the Course				
216U03E513	Database Management System				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Data Structure and Programming Logic

Course Objectives: This course provides a comprehensive understanding of database management systems. Students will learn core concepts, including data models, database architecture, and the role of DBMS in information systems. The course covers SQL for data definition, manipulation, and querying, as well as relational database design using ER models and normalization. Topics include database implementation, management, performance optimization, security, transaction management, concurrency control, and recovery.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand the role of DBMS in real-world applications and differentiate it from traditional file systems.
- CO2.** Understand various data models and apply ER modeling to design diagrams for real-world scenarios.
- CO3.** Understand the internal storage structures in a physical DB design
- CO4.** Understand indexing and hashing techniques, analyze query processing steps and cost, and apply query optimization strategies to improve database performance
- CO5.** Apply fundamental concepts of transaction processing, comparison with non-relational database and utility of it.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Introduction to Databases:		06	CO1
	1.1	Introduction, Purpose of Database Systems, DBMS system architecture , Data Models, Data independence		
	1.2	Database languages, Database Users and Administrator, Different types of Database Systems		
2	Database Models and SQL		09	CO2 CO3
	2.1	Database design phases, E-R Model, Constraints, E-R Diagrams, E-R design issues		
	2.2	Entity set, Extended E-R features		
	2.3	Relational model concepts, Constraints Relational Algebra: Unary, Binary and Set theory relational operations Data definition commands: attribute constraints, SET operations, Aggregate functions, Null Values, Nested sub queries, complex queries, Views Data control commands Data manipulation commands: Insert, Update, Joined relations Integrity and security: Domain constraints, Referential integrity, Triggers; Security and Authorization in SQL		
3	Relational Database Design		10	CO4
	3.1	Design guidelines for relational schemas, Functional Dependencies, Armstrong Axiom		
	3.2	First Normal form, Second Normal form, Third normal Form.		
	3.3	Decomposition and Desirable properties of decomposition, decomposition using multivalued dependencies, fourth normal form.		
	3.4	Overall database design		
4	Indexing, Hashing , Query processing and Optimization		10	CO4
	4.1	Basic concepts, ordered indices: dense and sparse, multilevel indices, secondary indices		
	4.2	Hashing: Static hashing, dynamic hashing, comparison of ordered indexing and hashing		
	4.3	Query processing: Steps involved in query processing, measures of query cost, algorithms for SELECT and PROJECT operations.		
	4.4	Query Optimization: Overview, Transformation of relational expressions, Estimating statistics, Choice of evaluation plan		

5	Transactions and Non-relational databases		10	CO5
	5.1	Transaction Concepts, Transaction state, ACID properties, concurrent executions.		
	5.2	Distributed Databases: Sharding and Replication, Consistency, Master Slave Replication, Peer to Peer replication, Introduction to cache, The CAP Theorem, NoSQL Data Models		
	5.3	NoSQL approach, Key-Value and Document Data Models, Column-Family Stores, Comparison of relational databases to new NoSQL stores, MongoDB, Cassandra		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Korth, Slberchatz, Sudarshan	<i>Database System Concepts</i>	McGraw Hill Education, India	7 th Edition, 2018
2	Raghu Ramakrishnan and Johannes Gehrke	<i>Database Management Systems</i>	McGraw Hill Education, India	3 rd Edition 2002
3	Elmasri and Navathe	Fundamentals of Database Systems	Pearson Education, India	7 th Edition, 2019
4	G.K. Gupta	<i>Database Management Systems</i>	McGraw Hill Education, India	6 th Edition, 2011